A3-PS013-TC537

Python-Based Incident Timeline and Evidence Collection Tool

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AIORI Hackathon 3
October 10, 2026

Problem Description

During a cybersecurity incident, investigation teams must manually examine logs from Linux servers, web applications, and other systems to understand what happened.

Problem Objective:

Develop a lightweight digital-forensics tool that processes multiple log formats, preserves original evidence, normalizes timestamps to UTC, generates cryptographic hashes, and creates a searchable incident timeline to help investigation teams reconstruct cyber incidents faster and more consistently.

Key Objectives

- **Fragmented Logs:** Logs are scattered across different systems and applications.
- **Different Log Formats:** Systems store events in different formats, making analysis difficult.
- **Timestamp Inconsistency**: Different time zones and clock differences complicate event ordering.
- **Time-Consuming Investigation:** Manually correlating events delays incident analysis.
- **Evidence Integrity:** Original logs must be preserved and protected against unnoticed modifications.
- **Lack of Unified Timeline:** Investigators lack a centralized chronological view to reconstruct cyber incidents efficiently.

Background

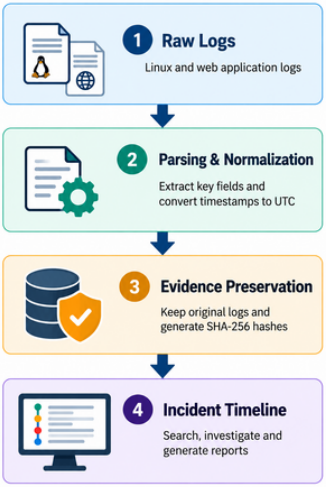
System logs contain important information about user activity, authentication attempts, application requests, and suspicious events. These records help investigators reconstruct the sequence of a cyber incident.

Log Analysis: Logs record system activity and suspicious events.

Key Challenges: Different formats and timestamps make analysis difficult.

Evidence Integrity: Preserve original logs and generate SHA-256 hashes.

Our Approach: Normalize timestamps and build a searchable incident timeline.



Literature & RFC Review

Relevant Standards and Research

1. NIST SP 800-86 — Guide to Integrating Forensic Techniques into Incident Response
Provides guidance on collecting, preserving, and analyzing digital evidence.

2. RFC 5424 — The Syslog Protocol
Defines a standard format for system log messages.

3. SHA-256 Hashing
Helps verify evidence integrity by detecting changes to collected log files.

4. Project Gap
Our tool combines log normalization, timestamp alignment, evidence hashing, and incident timeline generation in one workflow.

References

NIST SP 800-86: https://csrc.nist.gov/pubs/sp/800/86/final

RFC 5424: https://www.rfc-editor.org/rfc/rfc5424

Proposed Solution

Python-Based Incident Timeline and Evidence Collection Tool

The proposed system automates log analysis and evidence organization to help investigation teams reconstruct cyber incidents efficiently.

1. Log Ingestion: Import Linux authentication and web application logs.
2. Log Normalization: Extract key fields and convert timestamps to UTC.
3. Evidence Preservation: Retain original logs and generate SHA-256 hashes.
4. Timeline Generation: Display chronological events and export incident reports.

Expected Outcome: Faster incident triage with organized, searchable, and integrity-checked evidence.

Optimization Proposed

The proposed system improves incident investigation by automating log processing, standardizing timestamps, preserving evidence integrity, and simplifying event analysis.

Before Optimization

- Manual log analysis
- Different timestamp formats
- Difficult event correlation
- Manual evidence verification
- Time-consuming report preparation
- Difficult parsing error detection

After Optimization

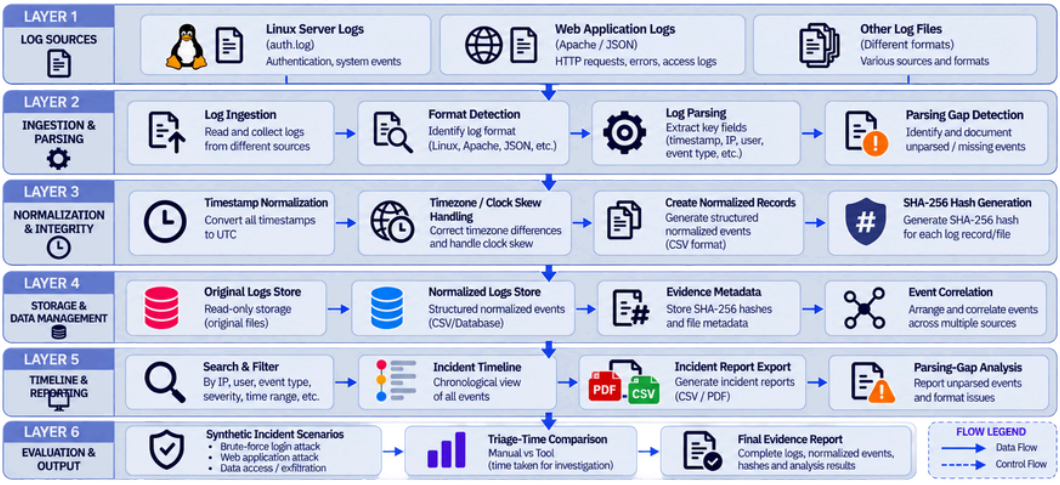
- Automated log processing using Python.
- Standardized UTC timestamps.
- Chronological incident timeline generation
- SHA-256 evidence integrity verification.
- Automated incident report generation.
- Documented parsing gaps and errors.

Expected Benefit: Reduced triage effort, improved evidence integrity, and faster incident investigation.

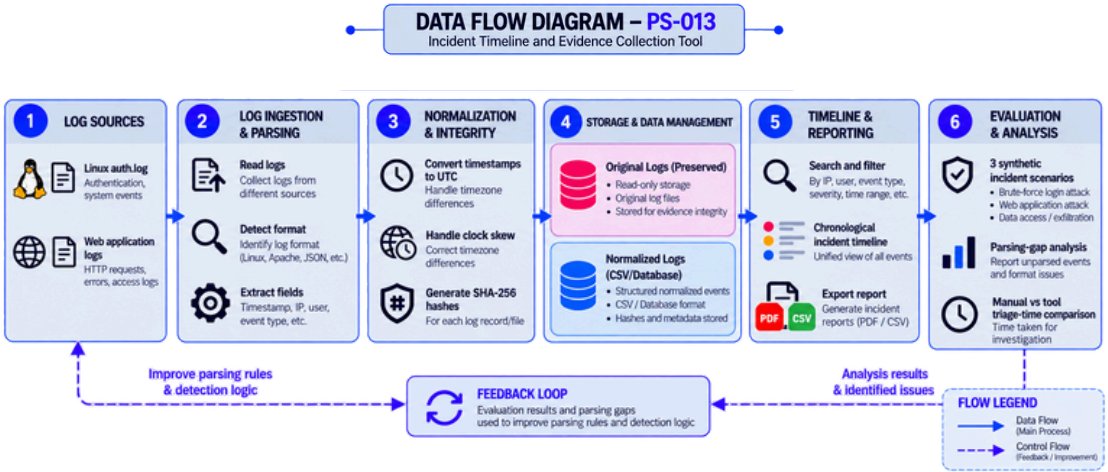
Solution Architecture & Design

System Architecture – PS-013:

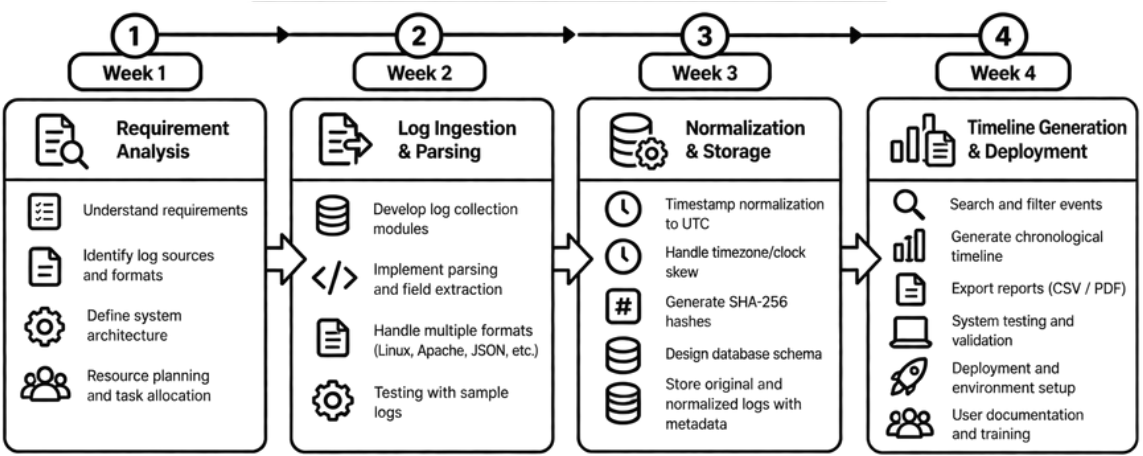
Incident Timeline and Evidence Collection Tool



Data Flow Diagram



Deployment Timeline



Bibliography I

[1] National Institute of Standards and Technology (NIST). Guide to Integrating Forensic Techniques into Incident Response. Special Publication 800-86, 2006. https://csrc.nist.gov/pubs/sp/800/86/final

[2] National Institute of Standards and Technology (NIST). Computer Security Incident Handling Guide. Special Publication 800-61 Revision 2, 2012. https://csrc.nist.gov/pubs/sp/800/61/r2/final

[3] National Institute of Standards and Technology (NIST). Secure Hash Standard (SHS). Federal Information Processing Standards Publication 180-4, 2015. https://csrc.nist.gov/pubs/fips/180-4/upd1/final

[4] AIORI Hackathon 3. PS-013: Incident Timeline and Evidence Collection for Small Teams. Official Problem Statement Document.

Thank You for Your Attention!